Formal Languages & Automata — Unit 4 Tutorial Sheet

Pushdown Automata

• Q1. Design a PDA that accepts { a^n b^2n : n ≥ 1 } by empty stack.
• Q2. Convert the grammar S → aSa | bSb | c into an equivalent PDA.
• Q3. Show that { a^n b^n c^n : n ≥ 0 } is not context-free using the pumping lemma for CFLs.
• Q4. Is every language accepted by a DPDA by final state also accepted by empty stack? Justify.